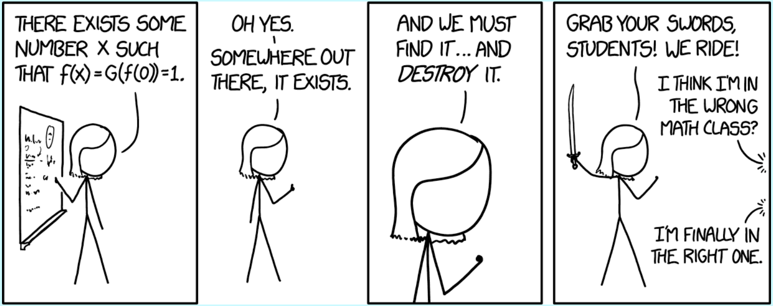
Quantifiers

Intro and Definitions



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Single-instance vs. universal statements

Question: What is the difference between these statements?

- If I'm at the pond right now, then I am throwing rocks
- Whenever I'm at the pond I throw rocks

How about this one?

- If I'm at the pond then I'm throwing rocks

Quantifiers

Quantifiers are defined with respect to a universe U. They either describe something that is true about *all* elements of U, or *at least one* element of U.

$\forall$ = "for all" (**universal quantifier**). Usage:

$$\forall x \in U, \text{ assertions about } x$$

$\exists$ = "there exists" (**existential quantifier**). Usage:

$$\exists x \in U, \text{ assertions about } x$$

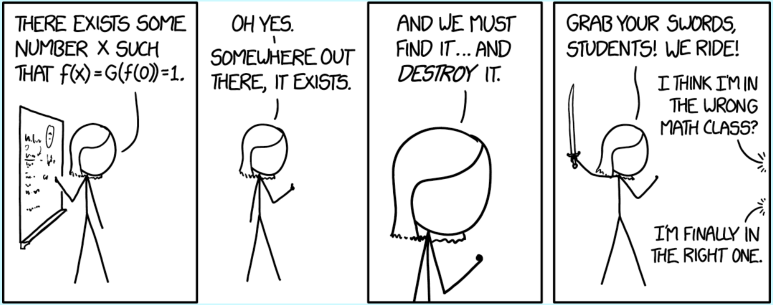
Examples

Here are examples of phrases (which may be true or false) that include the notions of $\forall$ and/or $\exists$:

1. All integers have a prime factor
2. There is a multiple of 7 whose digits are all 0s and 1s
3. Every natural number has a multiple whose digits are all 0s and 1s
4. For all integers a, b, c, if $a|b$ and $b|c$ then $a|c$
5. Any programming language that is Turing-complete can be used to solve any computational problem
6. There is a magical map which shows every creature alive in Hogwarts

Quantifiers

Single quantifiers



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Examples - Single quantifier

1. Let $F(x) =$ "x has a friend". Let the universe be $P = \{\text{people}\}$. Interpret the following:
   (a) $\forall x\, F(x)$

   (b) $\exists x\, F(x)$

Examples - Negation of single quantifier

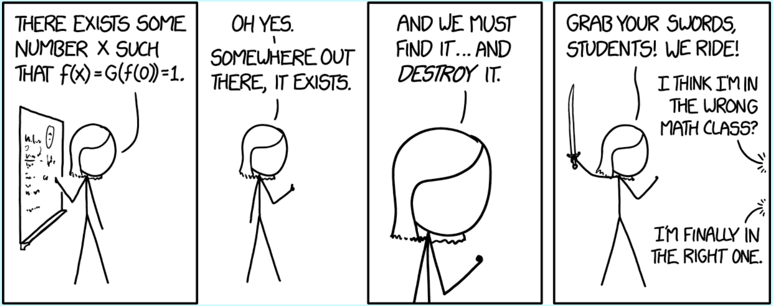
2. Now interpret the following negations in English, rewrite them in English so that the “not” is no longer in front, and convert them back into math notation.

   (a) $\neg \forall x \, F(x)$

   (b) $\neg \exists x \, F(x)$

Quantifiers

Double quantifiers



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Examples - Double quantifiers

3. Let $N(x, y) =$ "x knows y's name". Let the universe be $S =$ {students in this class}. Interpret the following:

   (a) $\forall x\, \exists y\, N(x, y)$

   (b) $\exists y\, \forall x\, N(x, y)$

   (c) $\forall y\, \exists x\, N(x, y)$

   (d) $\exists x\, \forall y\, N(x, y)$

   (e) $\forall x\, \forall y\, N(x, y)$

   (f) $\forall y\, \forall x\, N(x, y)$

   (g) $\exists x\, \exists y\, N(x, y)$

   (h) $\exists y\, \exists x\, N(x, y)$

Examples - Negation of double quantifiers

4. Now interpret the following negations in English, rewrite them in English so that the “not” is no longer in front, and convert them back into math notation.

   (a) $\neg\forall x\, \exists y\, N(x, y)$

   (b) $\neg\exists y\, \forall x\, N(x, y)$

   (c) $\neg\forall y\, \exists x\, N(x, y)$

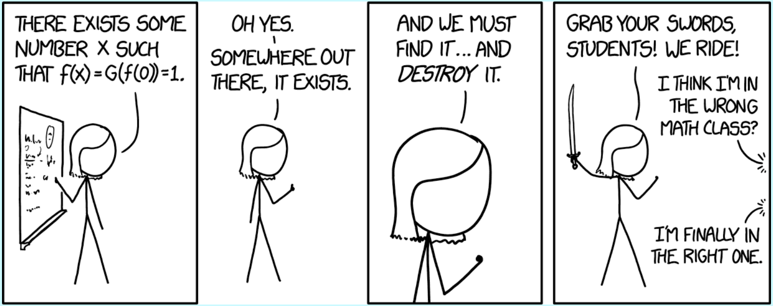
   (d) $\neg\exists x\, \forall y\, N(x, y)$

   (e) $\neg\forall x\, \forall y\, N(x, y)$

   (f) $\neg\exists x\, \exists y\, N(x, y)$

Quantifiers

Proof templates



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Proofs with single quantifiers

Claim
$\forall x \in U,\ p(x)$

Proof.
Let $x \in U$. (x could be anything in U)
...
(show $p(x)$ is true) $\square$

Claim
$\exists x \in U,\ p(x)$

Proof.
Let $x =$ (some specific element of U).
...
(show p(that elt) is true) $\square$

Examples:

Claim
$\forall x \in \mathbb{Z},\ x | 0$

Claim
$\exists x \in \mathbb{Z},\ x^2 = 49.$

Proofs with double quantifiers

Claim
$\forall x \in U, \exists y \in U,\ p(x, y)$

Proof.
Let $x \in U$. Let $y =$ (specific elt of U, can depend on x).
...
(show $p(x, y)$ is true) $\square$

Claim
$\exists y \in U, \forall x \in U,\ p(x, y)$

Proof.
Let $y =$ (some specific elt of U).
Let $x \in U$ (could be anything).
...
(show $p(x, y)$ is true) $\square$

Examples:

Claim
$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z},\ x < y$

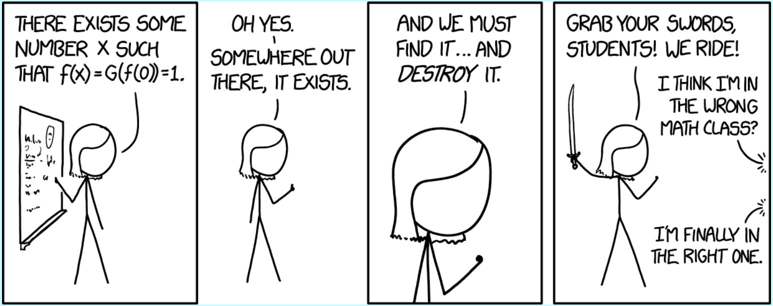
Claim
$\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z},\ x < y$

Disproving quantifiers

Thinking question: How would you disprove a quantified statement?

Further exploration

Existence statements can be very inspiring...



https://xkcd.com/1856/

Existence statements appear often in philosophy. Another xkcd comic references the example of Russell's teapot. Bertrand Russell put forth a hypothetical claim that there exists a teapot orbiting the sun between Earth and Mars. Russell points out that his claim, though unlikely, can never be refuted with absolute proof.